SUMS OF TWO SQUARES AND HEEGNER NORMS WITH FEW PRIME FACTORS, V: A PRIME PLUS A HEEGNER NORM (THE LINNIK PROBLEM FOR THE NINE DISCRIMINANTS OF CLASS NUMBER ONE)

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Abstract. Parts I–IV studied $N = n + (N - n)$ with both summands norms from an imaginary quadratic field of class number one, a sieve problem of dimension 2 whose remainder is a bilinear form in Kloosterman sums with one argument fixed. This part changes the problem to $N = p + N_D(x, y)$, a rational prime plus a norm from the field of discriminant $-D$, $D \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}$. For $D = 4$ this is the problem of Hardy and Littlewood solved by Linnik with the dispersion method. We prove, for all nine D by the divisor-switching argument of Friedlander and Iwaniec, the asymptotic formula $\sum_{p<N} r_D(N-p) = \frac{2\pi}{\sqrt{D}}\,\mathfrak{S}_D(N)\,\pi(N)+o(\pi(N))$ with the singular series in closed form, $\mathfrak{S}_D(N) = (1 + c_D(N)) \prod_{\ell\mid N}(1 - \chi_{-D}(\ell)/\ell) \prod_{\ell\nmid N}(1 + \chi_{-D}(\ell)/(\ell(\ell - 1)))$, where the factor at the ramified prime, $1 + c_D(N) = 1 - \chi_{-D}(N)/(D - 1)$ for odd D and 1 for $D = 4, 8$, comes from the second half of the hyperbola; the proof needs the Bombieri–Friedlander–Iwaniec level $\sqrt{N}(\log N)^A$ for the single residue class N, which is proved for fixed residues and which we state as a hypothesis, and for $D = 4$ the theorem is unconditional by Linnik. The sequence $\{N-p\}$ has sieve dimension 1, but the condition "$N-p$ is a norm" is not a sieve condition: the weighted sequence $r_D(N-p)$ has dimension 1 as well, and no level of distribution for it follows from the Bombieri–Vinogradov theorem, because the hyperbola puts the sieve modulus next to a modulus of size $\sqrt{N}$. So neither Chen's $N = p+P_2$ nor Richert's $N = p + P_3$ transfers to norms, and we say precisely what would be needed. Sifting the unweighted sequence by the inert primes with Iwaniec's half-dimensional sieve gives, unconditionally, $\gg N(\log N)^{-3/2}$ primes p with $N - p$ a norm up to at most two inert primes above $N^{1/2-\varepsilon}$; removing them is a Chen-type comparison of constants which the crude upper bound loses by an order of magnitude, and which we leave open. The reversed problem, $N = N_D(x, y) + P_3$, is unconditional by the lattice-point level $1/2$ and Richert's weighted sieve. Numerically the asymptotic formula holds to 1% for all nine D at $N \le 2 \cdot 10^5$, and for N even the proportion of primes $p < N$ with $N - p$ a norm with at most two prime factors is between 0.11 and 0.34.

1. Introduction and statement of results

1.1. **The problem.** Let $D \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}$, let $K = \mathbb{Q}(\sqrt{-D})$, $\chi = \chi_{-D}$ the Kronecker symbol of conductor D, w_D the number of units $(6, 4, 2, \dots)$, and N_D the principal form, $x^2 + y^2$, $x^2 + 2y^2$ or $x^2 + xy + \frac{D+1}{4}y^2$. Since $h(-D) = 1$, the number of representations of $n \ge 1$ by N_D is

$$r_D(n) = w_D \sum_{d\mid n} \chi(d) = w_D\, \lambda(n), \qquad \lambda = 1 * \chi,$$

and $r_D(n) > 0$ exactly when n is a norm, i.e. when every inert prime divides n to an even power. Parts I–IV [4, 5, 6, 7] treated $N = n + (N - n)$ with both summands norms, through the theta series $\theta_D(z)\theta_D(pz)$; the sieve there has dimension 2 and the remainder beyond the level $1/2$ is a bilinear form in Kloosterman sums with the single argument N fixed. Here we consider instead

$$N = p + N_D(x, y), \qquad p \text{ prime}, \quad (x, y) \in \mathbb{Z}^2,$$

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for a large integer N, through the sum

$$S_D(N)=\sum_{p<N} r_D(N-p)=w_D\sum_{p<N}\sum_{d\mid N-p}\chi(d).$$

For $D=4$ this is the Hardy–Littlewood problem $N=p+x^2+y^2$, which Linnik solved with the dispersion method [15, Chapter 7]; Hooley [11] had obtained the asymptotic under the extended Riemann hypothesis. For the shifted problem $p=x^2+y^2+1$, Bredihin [2] extended Linnik's argument, and Friedlander and Iwaniec [8] gave a short proof from the Bombieri–Friedlander–Iwaniec theorem [1] on primes in arithmetic progressions to moduli $\sqrt{x}(\log x)^A$. We follow the latter route, for all nine D at once and for the fixed-N problem, and we compute the singular series explicitly, including the factor at the ramified prime that the problem $N=p+N_D(x,y)$ has and the problem $p-1=x^2+y^2$ does not.

1.2. **The asymptotic formula.** The input is the following statement, which is the theorem of Bombieri, Friedlander and Iwaniec [1, Theorem] with the residue class $a=N$ in place of a fixed residue class.

Hypothesis 1.1 (H(N))**.** For every $A>0$, uniformly in $X\le N$,

$$\sum_{\substack{d\le\sqrt{N}(\log N)^A\\(d,N)=1}}\Big|\pi(X;d,N)-\frac{\pi(X)}{\varphi(d)}\Big|\ll_A N\Big(\frac{\log\log N}{\log N}\Big)^2.$$

For $X\le N(\log N)^{-3}$ the bound is trivial ($\pi(X;d,N)\le X/d+1$), so the content is in $N(\log N)^{-3}\le X\le N$. The Bombieri–Vinogradov theorem gives the bound with $d\le\sqrt{N}(\log N)^{-B}$ and the saving $(\log N)^{-A}$, uniformly in the residue class; the extension to $\sqrt{N}(\log N)^A$ is proved in [1] for a fixed residue class a, with the implied constant depending on a, and the uniform versions known to us (Maynard [16], Lichtman [14]) either restrict the moduli to those with a factor of a prescribed size or replace the absolute values by well-factorable weights. Hypothesis H(N) as stated is therefore open; it is what the divisor-switching proof needs, and nothing more.

Theorem 1.2. *Let $c_D(N)=\varphi(D)^{-1}\sum_{a \bmod D,\ (a,D)=1}\chi_{-D}(N-a)$, so that $c_D(N)=-\chi_{-D}(N)/(D-1)$ for odd D and $c_D(N)=0$ for $D=4,8$, and let*

$$\mathfrak{S}_D(N)=\big(1+c_D(N)\big)\prod_{\ell\mid N}\Big(1-\frac{\chi(\ell)}{\ell}\Big)\prod_{\ell\nmid N}\Big(1+\frac{\chi(\ell)}{\ell(\ell-1)}\Big).$$

(i) *If* H(N) *holds, then*

$$S_D(N)=\frac{2\pi}{\sqrt{D}}\,\mathfrak{S}_D(N)\,\pi(N)+O\Big(N\Big(\frac{\log\log N}{\log N}\Big)^2\Big).$$

(ii) *For $D=4$ the asymptotic $S_4(N)\sim\pi\,\mathfrak{S}_4(N)\,\pi(N)$ holds unconditionally, with the error term $O(N(\log N)^{-1-\eta})$ of Linnik's theorem* [15, Chapter 7].

In particular $\mathfrak{S}_D(N)\gg 1$ for every N (every factor is at least $1-\frac12$ at $\ell=2$ and $1-\frac{1}{D-1}$ at the ramified prime, and the product over $\ell\nmid N$ converges absolutely), so that, under H(N)*, every large N is a prime plus a norm from K, with $\gg N^{1-\varepsilon}$ representations.*

The constant is the mean value of r_D: $w_DL(1,\chi_{-D})=2\pi h/\sqrt{D}=2\pi/\sqrt{D}$ by the class number formula, and the singular series is the correction at each prime, with the two halves of the hyperbola contributing 1 and $c_D(N)$ respectively (Section 2). The factor $1+c_D(N)$ is the local density at the ramified prime: for $D=3$, $N_3(x,y)=x^2+xy+y^2\equiv(x-y)^2 \pmod 3$ never

is 2 mod 3, so for $N \equiv 1 \pmod 3$ the primes $p \equiv 2 \pmod 3$ are lost ($\frac{1}{2}$), for $N \equiv 2$ none are lost and the primes $p \equiv 2$ give $N - p \equiv 0$, which has twice the density of norms ($\frac{3}{2}$), and for $N \equiv 0$ the two effects cancel (1). Section 4 verifies the formula to 1% for all nine D and $N \le 2 \cdot 10^5$; without the factor $1 + c_D(N)$ the ratios for $D = 3$ are 0.50, 1.00, 1.50 according to N mod 3.

Remark 1.3. (a) The dispersion method of Linnik does not need H(N); it proves (ii) with a power of $\log N$ saved, and it is what makes the problem $N = p + x^2 + y^2$ unconditional. We expect it to apply to the forms N_D as it applies to $x^2 + y^2$ (the inputs are the divisor structure of r_D and the equidistribution of the norms in residue classes, both of which the class number one fields share), but we have not carried the dispersion argument out for $D \ne 4$, and we do not claim (ii) for them. (b) The error term in (i) is $o(\pi(N))$ and no better; this is the error term of [8], and it comes from the level $\sqrt{N}(\log N)^A$ with the saving $(\log\log N/\log N)^2$ of [1].

1.3. **The sieve.** The sequence $\mathcal{A} = \{N - p : \ p < N\}$ is the sequence of Chen's theorem: its sieve dimension is 1, its level of distribution is 1/2 by the Bombieri–Vinogradov theorem, and the weighted sieve of Richert gives $N = p + P_3$, Chen's switching $N = p + P_2$, for every large even N. None of this uses or produces the condition that $N - p$ be a norm. The relevant sequence is the weighted one, $a_m = r_D(m)$ for $m = N - p$, and its level of distribution is Theorem 1.2 in arithmetic progressions on average over the modulus.

Proposition 1.4. (i) *For $\mathcal{A} = \{N - p\}$ and a prime $q \nmid N$, $\#\{p < N : \ q \mid N - p\} = \pi(N)/(q-1) + O(E(N;q))$, so that $\omega(q) = q/(q-1)$, $\omega(q) = 0$ for $q \mid N$, and $\prod_{w \le q < z}(1 - \omega(q)/q)^{-1} = \frac{\log z}{\log w}(1 + O(1/\log w))$: dimension 1. The level of distribution is 1/2.*

(ii) *For the weighted sequence $a_m = r_D(m)1_{N-m \text{ prime}}$ and a prime $q \nmid DN$, the expected local density is $\omega_D(q) = (1 + \chi(q))\, q/(q-1)$: $2q/(q-1)$ at the split primes, 0 at the inert ones, and $\sum_{q<z} \omega_D(q)/q = \log\log z + O(1)$: dimension 1 on average, with the half-dimensional structure of the inert primes. The quantity whose average over $q \le N^\theta$ a sieve needs is*

$$\sum_{\substack{p<N \\ q \mid N-p}} r_D(N-p) - \frac{\omega_D(q)}{q}\,\frac{2\pi}{\sqrt{D}}\mathfrak{S}_D(N)\pi(N),$$

and by the hyperbola (Section 2) its control at the level N^θ is equivalent to the equidistribution of the primes $p \equiv N \pmod{qb}$ for $q \le N^\theta$, $b \le \sqrt{N}(\log N)^A$, with the weights $\chi(b)$: moduli of size $N^{1/2+\theta}(\log N)^A$ in the single residue class N. No $\theta > 0$ follows from the Bombieri–Vinogradov theorem, and Hypothesis H(N) *gives $\theta = 0$ only.*

(iii) *If the weighted sequence had the level $\theta = 1/2$ (in the form needed by the weighted linear sieve), Richert's weighted sieve would give $N = p + P_3$ with $N - p$ a norm, for every large admissible N; Chen's switching would need, in addition, the level 1/2 for the sequences $\{N - q_1q_2m\}$ with q_1, q_2 split primes in prescribed ranges, and would give $N = p + P_2$ with $N - p$ a norm. Neither is proved here. The unweighted sequence, sifted by the inert primes with the half-dimensional sieve, gives $\gg N(\log N)^{-3/2}$ primes p with $N - p$ a norm up to at most two inert primes above $N^{1/2-\varepsilon}$ (Theorem 3.1), and the removal of those inert primes is a Chen-type comparison of constants that is not closed here (Proposition 3.2).*

Admissible N. For $D \equiv 3 \pmod 8$ the prime 2 is inert, so an even value $N - p$ of N_D is divisible by 4, and $\Omega(N - p) \le 2$ forces $N - p = 4$: the statement "$N = p + P_2$ with $N - p$ a norm" can only hold for even N when $D \in \{3, 11, 19, 43, 67, 163\}$ (the data of Section 4 show the count 0 at every odd N for these D). For $D = 7$ (2 split) and $D = 4, 8$ (2 ramified) there is no parity condition.

The content of Proposition 1.4 is negative: the change of problem removes one of the two obstacles of Parts I–IV, the dimension 2 (the prime side contributes nothing to the dimension),

but not the other. The fixed argument N of the Kloosterman sums of Part III reappears as the fixed residue class N of the primes, at moduli qb with $b \asymp \sqrt{N}$, and this is the same square-root barrier in a different guise. What the change of problem does give, at the level $\sqrt{N}(\log N)^A$ of the single class N, is the asymptotic formula of Theorem 1.2, with every local factor explicit; and for $D = 4$ Linnik's theorem gives that unconditionally.

2. Proof of Theorem 1.2

2.1. **The singular series.**

Proposition 2.1. *For every $N \geq 1$, $\sum_{(d,N)=1} \chi(d)/\varphi(d) = L(1,\chi)\,\mathfrak{E}_D(N)$ with*

$$\mathfrak{E}_D(N) = \prod_{\ell \mid N}\Big(1 - \frac{\chi(\ell)}{\ell}\Big)\prod_{\ell \nmid N}\Big(1 + \frac{\chi(\ell)}{\ell(\ell-1)}\Big),$$

and $c_D(N) = \varphi(D)^{-1}\sum_{(a,D)=1}\chi(N-a)$ equals $-\chi(N)/(D-1)$ for odd D and 0 for $D = 4, 8$.

Proof. The sum is multiplicative in d; at a prime $\ell \nmid N$ the local factor is $1 + \sum_{j\geq 1}\chi(\ell)^j/\varphi(\ell^j) = 1 + \frac{\chi(\ell)}{\ell-1}\big(1 - \frac{\chi(\ell)}{\ell}\big)^{-1}$, and $\big(1 + \frac{\chi(\ell)\ell}{(\ell-1)(\ell-\chi(\ell))}\big)\big(1 - \frac{\chi(\ell)}{\ell}\big) = \frac{\ell^2-\ell+\chi(\ell)}{\ell(\ell-1)} = 1 + \frac{\chi(\ell)}{\ell(\ell-1)}$; at $\ell \mid N$ the factor is $1 = (1-\chi(\ell)/\ell)\cdot(1-\chi(\ell)/\ell)^{-1}$. Multiplying out gives $L(1,\chi)\mathfrak{E}_D(N)$. For odd D (a prime), $\sum_{a=1}^{D-1}\chi(N-a) = \sum_{b\not\equiv N}\chi(b) = -\chi(N)$ since $\sum_{b \bmod D}\chi(b) = 0$. For $D = 4$, $\chi_{-4}(N-1) + \chi_{-4}(N-3) = \chi_{-4}(N-1) + \chi_{-4}(N+1) = 0$ because $N - 1 \equiv -(N+1) \pmod 4$ and $\chi_{-4}(-1) = -1$. For $D = 8$ and N even, $\{N - a : \ a \text{ odd}\}$ is the set of odd residues modulo 8, on which χ_{-8} sums to 0; for N odd every $N - a$ is even and $\chi_{-8}(N-a) = 0$. □

2.2. **Divisor switching.**

For $m \geq 1$ write $m = D^k m'$ with $D \nmid m'$. Since $\chi(D^j) = 0$ for $j \geq 1$, $\lambda(m) = \lambda(m')$, and for $(e, D) = 1$, $\chi(m'/e) = \chi(m')\chi(e)$ whenever $e \mid m'$ (χ is real and completely multiplicative). For $y > 0$, splitting the divisors of m' at y and writing the large ones as m'/e,

$$\lambda(m) = \sum_{\substack{d\mid m'\\ d\leq y}}\chi(d) + \chi(m')\sum_{\substack{e\mid m'\\ e<m'/y}}\chi(e) = \sum_{\substack{d\mid m\\ d\leq y}}\chi(d) + \chi(m')\sum_{\substack{e\mid m'\\ e<m'/y}}\chi(e), \tag{1}$$

the first sum being the same over m and m' because $\chi(d) = 0$ unless $(d, D) = 1$. Let $w \geq 0$ be smooth, supported in $[1, 2]$, with $\int w(t)\,\mathrm{d}t/t = 1$, and $Y \geq 1$. Integrating (1) against $w(y/Y)\,\mathrm{d}y/y$ and substituting $y \mapsto m'/y$ in the second sum,

$$\lambda(m) = \int_0^\infty \Big[w\Big(\frac{y}{Y}\Big) + \chi(m')\,w\Big(\frac{m'}{yY}\Big)\Big]\sum_{\substack{b\mid m'\\ b<y}}\chi(b)\,\frac{\mathrm{d}y}{y}, \tag{2}$$

which is [8, (2.3)] with the factor $\chi(m')$ added. Take $Y = \sqrt{N}$. For $m = N - p \leq N$ the integrand vanishes unless $y \leq 2\sqrt{N}$, so all moduli b are below $2\sqrt{N}$.

2.3. **The first half.**

Summing the term $w(y/Y)$ of (2) over $p < N$ gives

$$\int_0^\infty w\Big(\frac{y}{Y}\Big)\sum_{b<y}\chi(b)\,\pi(N; b, N)\,\frac{\mathrm{d}y}{y}$$

(for b with $(b,N)>1$ the progression contains at most one prime, and these $b\le 2\sqrt{N}$ contribute $O(\sqrt{N})$). By $\mathrm{H}(N)$ with $X=N$,

$$\sum_{b<2\sqrt{N}}\Big|\pi(N;b,N)-\frac{\pi(N)}{\varphi(b)}\Big|\ll N\Big(\frac{\log\log N}{\log N}\Big)^2,\qquad \sum_{\substack{b<y\\(b,N)=1}}\frac{\chi(b)}{\varphi(b)}=L(1,\chi)\mathfrak{E}_D(N)+O(y^{-1+\varepsilon}),$$

the second by Proposition 2.1 and the convergence of $\sum\chi(b)/\varphi(b)$ (partial summation against $\sum_{b\le t}\chi(b)\ll D$). With $\int w(y/Y)\,\mathrm{d}y/y=1$, the first half is

$$L(1,\chi)\mathfrak{E}_D(N)\,\pi(N)+O\Big(N\Big(\frac{\log\log N}{\log N}\Big)^2\Big).$$

2.4. **The second half.** The term $\chi(m')w(m'/(yY))$ summed over $p<N$ with $b\mid m'$, i.e. $b\mid N-p$ (as $(b,D)=1$), is

$$T_b(y)=\sum_{k\ge0}\sum_{\substack{p<N,\ p\equiv N\ (b)\\ D^k\,\|\,N-p}}\chi\Big(\frac{N-p}{D^k}\Big)\,w\Big(\frac{N-p}{D^kyY}\Big).$$

If $D\mid N$, only $p=D$ has $D\mid N-p$, and the terms $k\ge1$ are $O(1)$. Let $D\nmid N$. The inner sum is over the primes in the classes $p\equiv N\pmod b$, $p\equiv N+D^ka\pmod{D^{k+1}}$, $a\not\equiv0\pmod D$, with the smooth weight $f_{k,y}(p)=w((N-p)/(D^kyY))$ and the value $\chi(-a)$; by the Chinese remainder theorem these are $\varphi(D)$ classes modulo bD^{k+1} (for $k=0$) or $D-1$ classes (for $k\ge1$, a ranging over $1,\dots,D-1$), all coprime to bD when $(b,N)=1$. The smooth-weighted count of primes in a class modulo $q=bD^{k+1}$ is $\frac{1}{\varphi(q)}\sum_p f_{k,y}(p)$ plus an error which, by partial summation in f as in [8, Lemma 3.1], is bounded by $\sup_{X\le N}|\pi(X;q,\cdot)-\pi(X)/\varphi(q)|$ times $O(1)$; summed over $b<2\sqrt{N}$ this is the hypothesis $\mathrm{H}(N)$ for the moduli $q=bD^{k+1}\le2\sqrt{N}D^{k+1}$, which lie below $\sqrt{N}(\log N)^A$ as long as $D^{k+1}\le(\log N)^{A-1}$. For the larger k, $D^k>(\log N)^{A-2}$, the trivial bound $\sum_b\#\{p:\ bD^k\mid N-p\}\ll\sum_{b<2\sqrt{N}}(N/(bD^k)+1)\ll N\log N/D^k+\sqrt{N}$ is $O(N/(\log N)^{A-3})$. The main terms: for $k=0$ the classes carry $\chi(N-a)$, a coprime to D, and $\sum_{(a,D)=1}\chi(N-a)/\varphi(bD)=c_D(N)/\varphi(b)$; for $k\ge1$ the classes carry $\chi(-a)$, $a=1,\dots,D-1$, and $\sum_{a=1}^{D-1}\chi(-a)=0$. Hence

$$T_b(y)=\frac{c_D(N)}{\varphi(b)}\sum_{p<N}w\Big(\frac{N-p}{yY}\Big)+E_b(y),\qquad \sum_{b<2\sqrt{N}}\sup_y|E_b(y)|\ll N\Big(\frac{\log\log N}{\log N}\Big)^2.$$

Summing $\chi(b)T_b(y)$ over $b<y$ and integrating:

$$\int_0^\infty\sum_{p<N}w\Big(\frac{N-p}{yY}\Big)\frac{\mathrm{d}y}{y}=\sum_{p<N}\int_0^\infty w(t)\,\frac{\mathrm{d}t}{t}=\pi(N)$$

(substitute $t=(N-p)/(yY)$; $N-p\ge1$), the b-sum is again $L(1,\chi)\mathfrak{E}_D(N)+O(y^{-1+\varepsilon})$ with $y\gg\sqrt{N}\,(N-p)/N$, and the range of y where $w((N-p)/(yY))\ne0$ with $N-p\le N^{1-\varepsilon}$ (small y, where the truncation error $y^{-1+\varepsilon}$ is not small) contains $O(N^{1-\varepsilon})$ primes. The second half is therefore $c_D(N)L(1,\chi)\mathfrak{E}_D(N)\pi(N)+O(N(\log\log N/\log N)^2)$.

2.5. **Conclusion.** Adding the two halves and multiplying by w_D, $S_D(N)=w_DL(1,\chi)(1+c_D(N))\mathfrak{E}_D(N)\pi(N)+O(N(\log\log N/\log N)^2)$, and $w_DL(1,\chi_{-D})=2\pi/\sqrt{D}$ by the class number formula $L(1,\chi_{-D})=2\pi h/(w_D\sqrt{D})$ with $h=1$. This is (i). Part (ii) is Linnik's theorem [15, Chapter 7], whose main term is $\pi A_0\prod_{\ell\mid N}(\dots)\,N/\log N$ with the same local factors; the identification of his constant with $\pi\mathfrak{S}_4(N)$ is the computation of Proposition 2.1 with $c_4=0$. □

Remark 2.2. Where $\mathrm{H}(N)$ enters. The Bombieri–Vinogradov theorem covers $b \le \sqrt{N}(\log N)^{-B}$ with the saving $(\log N)^{-A}$ and no dependence on the residue class. The window $\sqrt{N}(\log N)^{-B} < b < 2\sqrt{N}$ contributes trivially $\ll \sum_b \pi(N;b,N) \ll \pi(N)\sum_b 1/\varphi(b) \asymp \pi(N)\,B\log\log N$, which is larger than the main term, and there is no cancellation to use in it, because the error terms $\pi(X;b,N)-\pi(X)/\varphi(b)$ enter with absolute values. This window is exactly what [1] controls for a fixed class, and what the dispersion method controls for the class N in the case $D=4$.

3. The sieve on the unweighted sequence

Proposition 1.4 says what the norm condition costs when it is put into the weights. The alternative is to leave the sequence $\mathcal{A}=\{N-p\}$ unweighted, where the Bombieri–Vinogradov theorem gives the level $1/2$ without any hypothesis, and to impose the norm condition by sifting: m is a norm exactly when no inert prime divides m to an odd power, so one sifts $\mathcal{A}$ by the inert primes. This sifting set has density $\frac12$ among the primes, which is the setting of Iwaniec's half-dimensional sieve [13], whose sieving limit is 1: the lower-bound function $f_{1/2}(s)$ is positive for $s>1$, so with the level $N^{1/2}$ the inert primes can be sifted up to $N^{1/2-\varepsilon}$. This section records what that gives, which is less than a norm, and why.

Theorem 3.1. *Let $0<\varepsilon<\frac12$, $y=N^{1/2-\varepsilon}$, and let N be even if $D\equiv 3 \pmod 8$ (when 2 is inert) and arbitrary otherwise. For $N\ge N_0(D,\varepsilon)$,*

$$\#\{p<N:\ q\nmid N-p \text{ for every inert prime } q<y\} \ \ge\ c(D,\varepsilon)\,\frac{N}{(\log N)^{3/2}},$$

and every $N-p$ counted is of one of the forms n_1, n_1q_0, $n_1q_0q_1$ with n_1 a norm from K and $q_0,q_1\ge y$ inert primes. The same count is $\ll_D N(\log N)^{-3/2}$.

Proof. Take $\mathcal{A}=\{N-p:\ p<N\}$, $X=\pi(N)$, and the sifting set $\mathcal{P}=\{q<y:\ q \text{ inert},\ q\nmid N\}$. For square-free d composed of primes of $\mathcal{P}$, $|\mathcal{A}_d|=\pi(N;d,N)=X\,g(d)+r(d)$ with $g(d)=1/\varphi(d)$, so $\omega(q)=q/(q-1)$ for $q\in\mathcal{P}$; the inert primes dividing N divide $N-p$ only for $p=q$, so they can be left out of $\mathcal{P}$ at the cost of $O(\log N)$ elements, and if 2 is inert and N is even then $2\nmid N-p$ for all odd p (for odd N the sieve would be empty, which is the parity condition of Section 1). The remainder satisfies $\sum_{d\le\sqrt{N}(\log N)^{-B}}\mu^2(d)3^{\omega(d)}|r(d)|\ll_A N(\log N)^{-A}$ by the Bombieri–Vinogradov theorem (the divisor-type weight is absorbed by Cauchy–Schwarz and the trivial bound, as usual). The sifting density is one-half: by the prime number theorem in the classes modulo D, $\sum_{u\le q<w,\ q\in\mathcal{P}}\omega(q)\log q/q=\frac12\log(w/u)+O(1)$, whence $\prod_{u\le q<w,\ q\in\mathcal{P}}(1-\omega(q)/q)^{-1}\le(\log w/\log u)^{1/2}(1+K/\log u)$, which is the one-sided condition $\Omega(\frac12)$ of the β-sieve. Rosser's weights of dimension $\frac12$ and level $\sqrt{N}(\log N)^{-B}$ then give (Iwaniec [13]; the β-sieve with $\kappa=\frac12$, $\beta=1$, [9, Chapters 11 and 14])

$$S(\mathcal{A},\mathcal{P},y)\ \ge\ X\,V(y)\big(f_{1/2}(s)-o(1)\big),\qquad V(y)=\prod_{q\in\mathcal{P}}\Big(1-\frac{1}{q-1}\Big),$$

$$s=\frac{\log(\sqrt{N}(\log N)^{-B})}{\log y}=\frac{1}{1-2\varepsilon}+o(1)>1,$$

and $f_{1/2}(s)>0$ for $s>1$. By Mertens' theorem in the two classes of primes, $V(y)\asymp_D(\log y)^{-1/2}$, so the lower bound is $\asymp N(\log N)^{-3/2}$; the upper bound $S\le XV(y)(F_{1/2}(s)+o(1))$ gives the same order. If $m=N-p$ survives, every inert prime factor of m is at least y, and since $m<N<y^3$ there are at most two of them counted with multiplicity: $m=n_1$, n_1q_0, $n_1q_0^2$ or $n_1q_0q_1$ with n_1 free of inert primes, i.e. a norm, and $n_1q_0^2$ is a norm as well. □

This is as far as the sieve on $\{N-p\}$ goes by itself: it produces $\gg N(\log N)^{-3/2}$ primes p with $N-p$ a norm up to at most two inert primes above $N^{1/2-\varepsilon}$, and it does not decide whether the inert primes are there. For $N-p$ to be a norm one has to remove the p with $N-p=n_1q_0$ or $n_1q_0q_1$, which is where Chen's switching would enter: one counts the pairs (q_0,n_1) with $N-q_0n_1$ prime instead of the primes p. The next proposition says what the sizes are.

Proposition 3.2. *Let $y=N^\theta$, $0<\theta<\frac12$, and let $E_\theta(N)$ be the number of $p<N$ with $N-p=n_1q_0$ or $n_1q_0q_1$, n_1 a norm free of inert primes below y, $q_0,q_1\ge y$ inert. Then $E_\theta(N)\ll_{D,\theta} N(\log N)^{-3/2}$, of the same order as the lower bound of Theorem 3.1 with $y=N^\theta$; more precisely, the Selberg upper-bound sieve applied to the primes $q_0\le N/n_1$ with $N-q_0n_1$ prime gives*

$$E_\theta(N)\le (c_1+o(1))\,\frac{N}{(\log N)^{3/2}},\qquad c_1\asymp\frac{16\,C_2\,c_K}{\sqrt\theta}\Big(\frac1\theta-1\Big),$$

with $C_2=\prod_{q>2}(1-(q-1)^{-2})=0.66$ and c_K the constant of $\prod_{q<y\ \mathrm{inert}}(1-1/q)\sim c_K(\log y)^{-1/2}$, while Theorem 3.1 gives the lower bound $(c_K\theta^{-1/2}f_{1/2}(1/2\theta)+o(1))\,N(\log N)^{-3/2}$ for the survivors. Since $f_{1/2}\le1$ and $16C_2(1/\theta-1)\ge10$ for $\theta\le\frac12$, the comparison fails by at least an order of magnitude for every θ.

Proof. For fixed $n_1\le N^{1-\theta}$ the primes $q_0\le N/n_1$ with $N-q_0n_1$ prime number at most $(8+o(1))\,\mathfrak S(n_1N)\,(N/n_1)/\log^2(N/n_1)$ by the Selberg sieve for the two linear forms q_0 and $N-n_1q_0$, with $\mathfrak S(n_1N)=2C_2\prod_{q\mid n_1N,\ q>2}\frac{q-1}{q-2}$. Summing over the n_1 in the dyadic ranges $n_1\asymp N^v$, $0\le v\le1-\theta$, with $\sum_{n_1\le Y}1/n_1$ over the n_1 free of inert primes below y being $\asymp c_K(\log y)^{-1/2}\log Y$ for $Y\ge y$ (the sifted integers have density $\prod_{q<y\ \mathrm{inert}}(1-1/q)$ by the fundamental lemma, as $Y\ge y$), the sum is $\asymp 16C_2c_K\,\theta^{-1/2}(\log N)^{-3/2}N\int_0^{1-\theta}(1-v)^{-2}\,\mathrm dv$, and $\int_0^{1-\theta}(1-v)^{-2}\,\mathrm dv=1/\theta-1$. The case $n_1q_0q_1$ is contained in the same count (q_1 absorbed into the cofactor). The lower bound is Theorem 3.1 with $s=\log\sqrt N/\log y=1/2\theta$ and $V(y)\sim c_K\theta^{-1/2}(\log N)^{-1/2}$ up to the factor $\prod(1-\frac{1}{q-1})/\prod(1-\frac1q)$, which is absorbed in c_K. □

Chen's theorem $N=p+P_2$ closes the analogous comparison, $f_1(\dots)-\frac12\int F_1(\dots)>0$, with a margin of a few percent, by applying the switching to the sieve weights themselves rather than to the whole exceptional set, and by using the linear-sieve upper bound F_1 in place of the constant 8. Whether the half-dimensional analogue closes is the question that decides "$N=p+P_k$ with $N-p$ a norm"; the crude bound of Proposition 3.2 says that it does not close without that refinement, and we have not carried the refinement out. We also note what the task of removing q_0 would give if it succeeded: the survivors of Theorem 3.1 have no restriction on the split primes, so $\Omega(N-p)$ is unbounded; to bound it one sifts the small split primes as well, which lowers the dimension from $\frac12$ only if z is sub-polynomial (the fundamental lemma with $z=\exp((\log N)^a)$, $a<1$, costs a factor $(\log N)^{-(1+a)/2}$ and bounds no Ω), or raises it to 1 on $[2,z)$ if $z=N^\delta$, where the linear sieve's limit $\beta_1=2$ forces $y\le N^{1/4}$ and admits three large inert primes. So the honest form of a theorem in this direction is "$N=p+m$, m a norm with $\Omega(m)\le k$" with a k that comes out of a weighted half-dimensional sieve with switching, and no such k is established here.

Proposition 3.3 (the reversed problem)**.** *For every $N\ge N_0(D)$ such that $N_D(x,y)\not\equiv N \pmod q$ has a solution for every prime q (which holds for all N), $N=N_D(x,y)+P_3$ has a solution: a norm plus an integer with at most three prime factors.*

Proof. Sift $\mathcal B=\{N-N_D(x,y):\ (x,y)\in\mathbb Z^2,\ 0<N_D(x,y)<N\}$, $|\mathcal B|\sim\frac{2\pi}{\sqrt D}N$. For $d\ge1$, $|\mathcal B_d|=\sum_{a \bmod d,\ N_D(a)\equiv N}\#\{(x,y)\equiv a\ (d):\ N_D(x,y)<N\}=\rho_N(d)\big(\frac{2\pi N}{\sqrt D\,d^2}+O(\frac{\sqrt N}{d}+1)\big)$, where

$\rho_N(d)$, the number of solutions of $N_D(x,y) \equiv N \pmod d$, is $d - \chi(d)$ for primes $d \nmid DN$ and $\ll d$ always; so $\omega(q) = \rho_N(q)/q = 1 - \chi(q)/q + O(1/q)$, the dimension is 1, and the remainders $r(d) \ll \sqrt{N} + d$ satisfy $\sum_{d \le N^{1/2-\varepsilon}} |r(d)| \ll N^{1-\varepsilon}$: the level is $\frac{1}{2}$, from the boundary of the ellipse alone. Richert's weighted sieve [10, Chapter 9], with the parameters of its application to $N = p + P_3$ [17] (elements $\le N$, level $N^{1/2-\varepsilon}$, dimension 1), gives $\gg N(\log N)^{-2}$ elements of $\mathcal{B}$ with at most three prime factors. □

For $D = 4$ this is weaker than Linnik's theorem, which gives $N = p + x^2 + y^2$ with p prime; for the other D it is an unconditional existence statement that does not need Hypothesis H(N), at the price of P_3 in place of p. It is the reversed problem, with the almost-prime on the rational side and the norm unrestricted, and it says nothing about $N = p +$ (norm with few prime factors). Chen's switching for $\mathcal{B}$ would need bilinear estimates for the lattice points, which Iwaniec's level $N^{1-\varepsilon}$ for $\{\varphi(x,y) + A\}$ [12] may supply; we have not pursued it.

4. Numerical verification

The script `linnik_heegner.py` computes $S_D(N)$ exactly (the table of r_D by the divisor sum, the primes by a sieve), $\mathfrak{S}_D(N)$ by the Euler product over $\ell \le 2 \cdot 10^6$ (tail $O(10^{-6})$), and the counts of primes $p < N$ with $N - p$ a norm and with $N - p$ a norm with $\Omega(N-p) \le 2$, for eleven values $10^4 \le N \le 2 \cdot 10^5$ (even, odd, prime, 3^9, 2^{16}, $60060 = 2^2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$) and all nine D; and the sieve of Section 3 at $N = 2 \cdot 10^5$ and $N = 100003$ for six D, with the survivors factored. Table 1 gives the extremes of the first computation, Table 2 the second.

D	$2\pi/\sqrt{D}$	$1 + c_D(N)$	$S_D(N)$	main term	ratio	ratio, 11 values of N	norm/π	norm, $\Omega \le 2$, /π
3	3.6276	1.5	179946	179317.7	1.0035	[0.9975, 1.0192]	0.500	0.340
4	3.1416	1	36700	36639.8	1.0016	[0.9798, 1.0233]	0.178	0.126
7	2.3748	7/6	24814	24870.0	0.9977	[0.9905, 1.0132]	0.244	0.166
8	2.2214	1	55494	54859.3	1.0116	[0.9624, 1.0139]	0.301	0.187
11	1.8945	0.9	41198	41450.8	0.9939	[0.9741, 1.0195]	0.259	0.164
19	1.4415	17/18	25004	25181.6	0.9929	[0.9646, 1.0394]	0.209	0.138
43	0.9582	43/42	26338	26328.1	1.0004	[0.9516, 1.0170]	0.223	0.148
67	0.7676	67/66	20364	20386.9	0.9989	[0.9505, 1.0232]	0.195	0.136
163	0.4921	163/162	12798	12705.7	1.0073	[0.9724, 1.0406]	0.149	0.109

Table 1. $S_D(N) = \sum_{p<N} r_D(N-p)$ against $\frac{2\pi}{\sqrt{D}}\mathfrak{S}_D(N)\pi(N)$ at $N = 2 \cdot 10^5$ ($\pi(N) = 17984$), the range of the ratio over the eleven N of the script, and the proportions of primes $p < 2 \cdot 10^5$ with $N - p$ a norm, and a norm with at most two prime factors. The ratios deviate most at $N \approx 10^4$ and for the larger D, where $\mathfrak{S}_D(N)\pi(N)$ is smallest. Script `linnik_heegner.py`, file `results/linnik_heegner.txt`; exact integer counts, floating point in the Euler products.

Table 2 is the sieve of Theorem 3.1 at $N = 2 \cdot 10^5$: the survivors of the inert-prime sieve up to $\sqrt{N}$ number $V\pi(N)$ to within 1–4% (the sifting range reaches the level, so the sieve functions are not yet in the asymptotic regime; the ratios exceed 1 where the inert primes near $\sqrt{N}$ are sparse), and every survivor is n_1 or $n_1 q_0$, as the theorem says. The exceptional part $n_1 q_0$ is between 21% ($D = 3$) and 51% ($D = 11$) of the survivors, i.e. of the same order as the norms, which is Proposition 3.2 in the data; it is not an order of magnitude larger, so the crude constant c_1 of that proposition overestimates the exceptional set, and the comparison that would remove q_0 may well close with the right weights. With the primes below 100 sifted as well (right-hand

D	inert primes $< \sqrt{N}$ sifted survivors	$/\pi(N)$	V	ratio	survivors norms	n_1q_0	all primes < 100 and inert primes $< \sqrt{N}$ sifted survivors	norms ($= P_2$)	n_1q_0
3	11145	0.6197	0.5955	1.041	8764	2381	3297	3297	0
4	4537	0.2523	0.2517	1.002	2373	2164	3218	1639	1579
7	5460	0.3036	0.3006	1.010	3333	2127	3278	1982	1296
8	9481	0.5272	0.5202	1.013	5142	4339	3207	1652	1555
11	9067	0.5042	0.5023	1.004	4400	4667	3185	1426	1759
163	3722	0.2070	0.2080	0.995	1950	1772	3144	1650	1494

TABLE 2. The sieve of Section 3 at $N = 2 \cdot 10^5$, $\pi(N) = 17984$, $\sqrt{N} = 447$. Left: the primes $p < N$ with $N - p$ free of inert primes below $\sqrt{N}$, their proportion against $V = \prod_{q<\sqrt{N} \text{ inert},\ q\nmid N}(1 - \frac{1}{q-1})$, and their structure, norms n_1 against n_1q_0 with one inert prime $q_0 \geq \sqrt{N}$ ($n_1q_0q_1$ does not occur at $y = \sqrt{N}$). Right: the same with the primes below 100 sifted as well, where every surviving norm has $\Omega \leq 2$ ($100^3 > N$). For $D = 3$ with $y = \sqrt{N}$ the survivors with $N - p = n_1q_0$ are 21%; for the other D between 39% and 51%. Script `linnik_heegner.py`; exact counts.

columns), the surviving norms are P_2's, between 1426 and 3297 of the 17984 primes, against between 0 ($D = 3$) and 1759 of the form n_1q_0.

Three features of the data. (a) The factor $1 + c_D(N)$ is real: for $D = 3$ the ratios without it are 0.508, 1.499, 1.008, 1.501, 1.496, 1.006, 0.509, 0.502, 0.510, 1.002, 1.505 for the eleven N, i.e. $\frac{1}{2}, \frac{3}{2}, 1$ according to $N \equiv 1, 2, 0 \pmod 3$; for $D = 7$ they are 0.836 and 1.156, i.e. $1 \mp \frac{1}{6}$; for $D = 4$ and 8 they are already 1. (b) At odd N and $D \equiv 3 \pmod 8$ the count of p with $N - p$ a norm with $\Omega \leq 2$ is 0 or 1 (the parity condition of Section 1); at even N it is between $0.11\,\pi(N)$ ($D = 163$) and $0.34\,\pi(N)$ ($D = 3$), against 0.13–0.50 for norms with any number of prime factors. (c) The sieve densities of $\{N - p\}$ at $N = 2 \cdot 10^5$, $D = 4$: the proportion of p with $(N-p, P(z)) = 1$ is 0.4160, 0.2796, 0.2112 for $z = 10, 30, 100$ against $\prod_{q<z,\, q\nmid N}(1-\frac{1}{q-1}) = 0.4167$, 0.2802, 0.2122 (dimension 1), and the proportion with no inert prime factor below y is 0.2910, 0.2412, 0.2100 for $y = 10^2, 10^3, 10^4$ against the half-dimensional product 0.2903, 0.2362, 0.2048; at $z = 10^3$ and $y = 10^5$ the sifting range reaches $\sqrt{N}$ and the products no longer describe the counts (1.098 and 0.926), which is the sieve limit, not a defect of the data.

5. What is proved and what is not

(1) The asymptotic formula for $S_D(N)$ with the explicit singular series, for all nine D, under Hypothesis H(N) (Theorem 1.2(i)); unconditionally for $D = 4$ (Linnik). The new content is the uniform treatment, the ramified factor $1 + c_D(N)$, and the identification of the exact input, a level $\sqrt{N}(\log N)^A$ for the single residue class N.

(2) The sieve dimension of $\{N - p\}$ is 1 (trivial, and not the point); the dimension of the norm-weighted sequence is 1 on average with a half-dimensional inert part; its level of distribution is not known, and in particular it is not given by the Bombieri–Vinogradov theorem (Proposition 1.4(ii)). Consequently no statement "$N = p + P_k$ with $N - p$ a norm" is proved here for any k, and the lower bound that a Chen-type argument would produce, $f_1(5) - \frac{1}{2}\int F_1(\dots)$ in some form, has no meaning until the level is established; we have not written such a constant.

(3) Unconditionally, the half-dimensional sieve on the unweighted $\{N - p\}$ gives $\gg N(\log N)^{-3/2}$ primes p with $N - p$ a norm up to at most two inert primes above $N^{1/2-\varepsilon}$ (Theorem 3.1); the exceptional set is of the same order, and the Chen-type

comparison that would remove it is not closed (Proposition 3.2): no "$N = p + P_k$ with $N - p$ a norm" is proved for any k. The reversed statement $N = N_D(x, y) + P_3$ is unconditional (Proposition 3.3) and, for $D = 4$, weaker than Linnik's theorem.

(4) What remains new in the sieve direction is the shape of the problem: one obstacle of Parts I–IV, the dimension 2, is gone; the other, the fixed N, is not, and it now sits in the residue class rather than in the Kloosterman argument. The natural next steps are (a) the dispersion method for N_D, $D \neq 4$, which would make Theorem 1.2 unconditional, and (b) the weighted half-dimensional sieve with switching for the inert primes above $N^{1/2-\varepsilon}$, i.e. the constants of Proposition 3.2 done properly, which is the analogue of Chen's computation and is open.

Data and code

The script `linnik_heegner.py` (Section 4; Python 3.12, numpy; about two minutes; exact integer counts) with its output `results/linnik_heegner.txt` is in the directory `two-squares-part5` alongside Parts I–IV. All computations are evidence, not proof; the theorems do not depend on them.

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